# Method families impose different evidence obligations

Descriptive workflow roles only. Hybrid workflows remain interpretive; no family maturity count is shown.

| | Data | Physics | Transfer | Uncertainty | Monitoring | Decision |
|---|---|---|---|---|---|---|
| **Unconstrained data-driven predictors** | Labelled cases with split provenance | Decision variable: mass, plume, pressure | Held-out geology, sensor or schedule | Reliability near decision thresholds | Observation and sensor physics | Threshold, rank or alarm cost |
| **Physics-guided and physics-informed models** | States, equations and valid regimes | Held-out residuals and coupling checks | Boundary and constitutive shifts | Parameters, physics and discrepancy | State-to-sensor relationship | Protected physical limit or action |
| **Neural operators and multi-fidelity surrogates** | Broad simulator ensembles | Mass, plume edge and pressure limits | Grid, geology, schedule and fidelity | Emulator discrepancy propagated downstream | Observation operator and model update | Reference-simulator decision comparison |
| **Graph models for connectivity and faults** | Topology and edge properties | Connectivity perturbation | Unseen graph or fault topology | Graph structure and edge values | Node or edge state updates | Fault-sensitive monitoring or action |
| **Probabilistic and generative inversion** | Prior ensemble and observation model | Posterior predictive checks | Prior, noise, sensor or geology shift | Calibration, priors and discrepancy | Assimilation timing and observation physics | Posterior changes a named decision |
| **Control and reinforcement-learning workflows** | Validated environment and constraints | Independent high-fidelity resimulation | Disturbance, failure and boundary stress | Risk-sensitive policy evaluation | Delayed, noisy or missing states | Constraint violations and safe fallback |
| **Digital-twin or monitoring-updated systems** | Versioned models and monitoring streams | Post-update physical checks | Sensor, phase and site changes | Before and after each update | Observation changes state or decision | Versioned update, decision and audit |

*Interpretation: architecture changes what must be demonstrated, but it does not replace decision-specific validation.*